

BFUTR 2026 INNOVATION MARKETPLACE & TECH SHOWCASE

ABSTRACT

Session Format: Innovation Showcase / Peer Lightning Presentation (15–20 Minutes)

Theme: Global Tech Diversity, Resilient Software Architectures, Emerging Market Innovation

Presenter: Fuad Ahmed Tuba, B.Sc. Computer Science (St. Mary's University)

Affiliation: Senior Software & Systems Engineer, Aduutech (Addis Ababa, Ethiopia)

Presentation Title

Architecting Resilient and Secure Software Systems in Emerging Markets: Engineering Lessons from Public Sector Infrastructure and Commercial Enterprise in East Africa

Executive Summary & Abstract

As global technology expands, software engineers in emerging African economies face unique systemic constraints: intermittent network bandwidth, fluctuating power availability, legacy hardware ecosystems, and critical cybersecurity vulnerabilities. Successfully delivering high-performance digital services under these conditions demands architectural resilience, lean software engineering, and defensive systems design.

Drawing upon practical field experience architecting database security for the Ethiopian Ministry of Revenue, developing client-facing web applications at Aduutech, and maintaining public school IT infrastructure for the Kolfe Keranio Education Bureau, this presentation showcases actionable engineering patterns for building fault-tolerant software systems in resource-constrained environments.

The presentation covers three core pillars:

- Resilient Data Architecture:** Strategies for implementing offline-first data synchronization, append-only relational ledgers, and transaction integrity that withstand unexpected network latency and connectivity outages.
- Defensive Systems Hardening on Minimal Footprints:** Hardening POSIX Linux environments, optimizing lightweight database instances (PostgreSQL/SQLite), and enforcing granular Role-Based Access Control (RBAC) without expensive enterprise proprietary software.
- Community-Centric Engineering:** How African software engineers can bridge the digital divide by pairing commercial software innovation with voluntary public-sector technology maintenance, creating sustainable local digital ecosystems.

Target Audience & Key Takeaways

- Audience:** Software Engineers, Systems Architects, Tech Founders, and Tech Leaders interested in global technology resilience and emerging market software scalability.
- Key Takeaway:** Attendees will discover pragmatic architectural principles that prove that robust, secure, and user-centric software can thrive under the most demanding infrastructural environments

worldwide.